Chapter10.sql

```sql
-- Tutorial 10.3: hours per standard hour, by labor type
SELECT h.WorkYear, h.LaborType, ROUND(h.Hours, 0) AS Hours,
    ROUND(o.StandardHours, 0) AS StandardHours,
    ROUND(h.Hours / o.StandardHours, 2) AS HoursPerStandardHour
FROM (
    SELECT SUBSTR(lt.WorkDate, 1, 4) AS WorkYear, lt.LaborType,
        SUM(lt.RegularHours + lt.OvertimeHours) AS Hours
    FROM LaborTimeEntry AS lt
    WHERE lt.LaborType <> 'NonManufacturing'
    GROUP BY WorkYear, lt.LaborType
) AS h
    INNER JOIN (
        SELECT SUBSTR(pc.CompletionDate, 1, 4) AS WorkYear,
            SUM(pcl.QuantityCompleted
                * i.StandardLaborHoursPerUnit) AS StandardHours
        FROM ProductionCompletionLine AS pcl
            INNER JOIN ProductionCompletion AS pc
                ON pc.ProductionCompletionID = pcl.ProductionCompletionID
            INNER JOIN Item AS i ON i.ItemID = pcl.ItemID
        WHERE pc.CompletionDate
            <= (SELECT MAX(WorkDate) FROM LaborTimeEntry)
        GROUP BY WorkYear
    ) AS o ON o.WorkYear = h.WorkYear
ORDER BY h.WorkYear, h.LaborType;
```

	WorkYear	LaborType	Hours	StandardHours	HoursPerStandardHour
1	2024	Direct Manufacturing	60449.0	70871.0	0.85
2	2024	Indirect Manufacturing	23188.0	70871.0	0.33
3	2025	Direct Manufacturing	57575.0	66109.0	0.87
4	2025	Indirect Manufacturing	35815.0	66109.0	0.54
5	2026	Direct Manufacturing	62764.0	66998.0	0.94
6	2026	Indirect Manufacturing	38473.0	66998.0	0.57

Execution finished without errors.
Result: 6 rows returned
At line 204:
SELECT h.WorkYear, h.LaborType, ROUND(h.Hours, 0) AS Hours,

Outlined: hours recorded per standard hour of output. Direct time stays below the standard in each year, while indirect time grows.